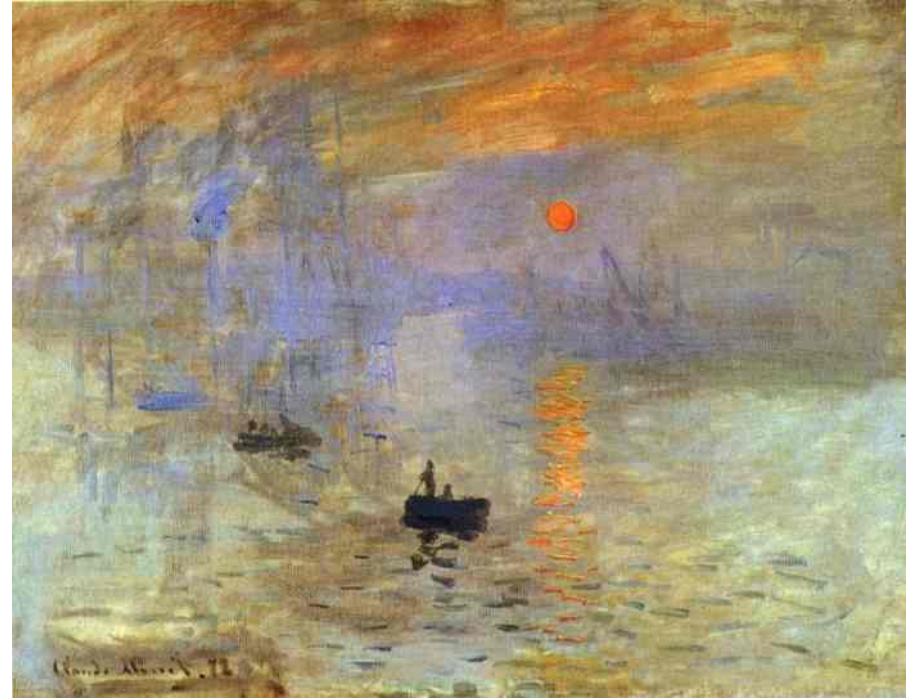


Human Vision and Computer Graphics

Historical Perspective

Human Visual Perception in Art

- Traditional Artists:
 - Little knowledge of inner workings of human visual system.
 - Able to create sense of depth and motion on flat static surfaces.
 - They view the human visual system as a black box. Through experimentation, they learned to exploit its features.
 - In a sense, they have reverse engineered the human visual system to learn what type of inputs elicit certain responses in the brain.



Monet (1872), *Impression Sunrise*

Knowledge About Human Visual System

- We now have a much better understanding of inner workings of human visual system:
 - Extensive research from various fields of science:
 - Physics
 - Biology
 - Psychology
 - Physiology
 - Neuroscience
 - In-depth studies of visual system of macaque monkey, which is closely related to that of humans.



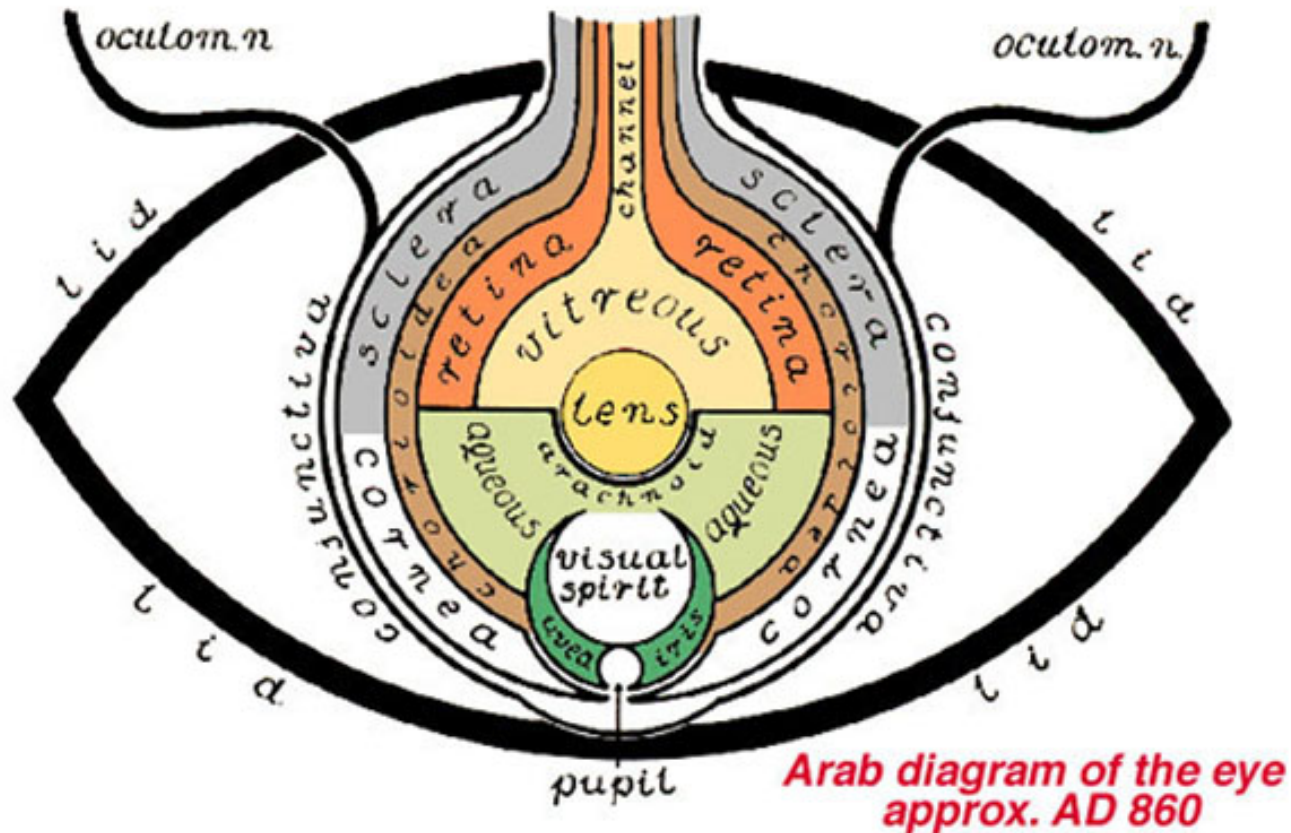
Knowledge About Human Visual System

- The visual system is far from being fully understood, but the knowledge gained thus far is quite substantial:
 - Especially of the early stages of the visual pathway.

Knowledge About Human Visual System

- The oldest theories about vision, light, and the eye date back to the early Greek philosophers.
 - Empedocles (490 - 430 BC)
 - Plato (428 - 348 BC)
 - Aristotle (384 - 322 BC)
 - Ptolemy (367 - 283 BC)
 - Euclid (325 - 265 BC)
- Their speculation about vision tended to be egocentric.
- They believed that the eyes produced rays which interacted with objects in the environment to facilitate vision.
- These theories persisted for several centuries.

Knowledge About Human Visual System



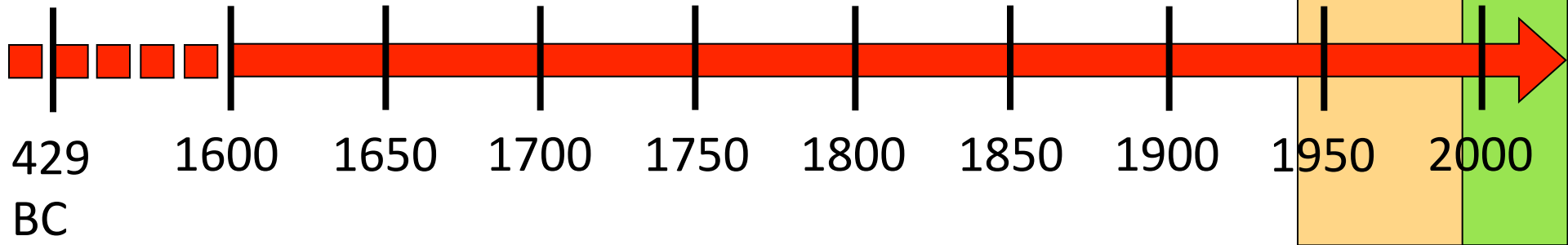
Earliest known diagram of the eye shows an area designated as the visual spirit.

Knowledge About Human Visual System

- Scientific study of vision did not begin until the start of the Enlightenment Period in the early 17th century.
- Why did it take so long?
 - Speculative and egocentric theories did not encourage scientific study.
 - Much of Europe was ravaged by poverty, plagues, and religious wars during the Middle Ages.
 - 5th century - 15th century

Historical Perspective

- Specialties of the early computer pioneers:
 - ◆ 1946's first program: Advantus computer (NAIC)
 - ◆ 1950's for experimental head-mounted display systems and eye tracking technology.
 - ◆ 1950's for some experiments in graphics science
 - ◆ 1961's first video game (Spacewars)





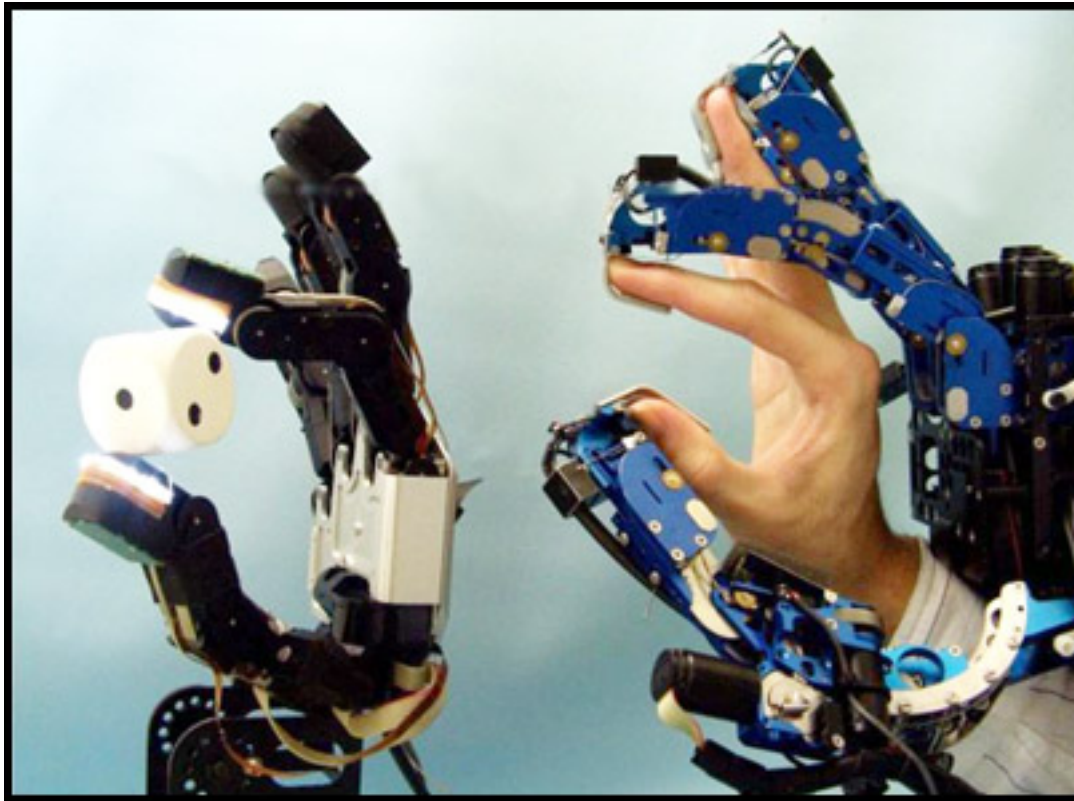
Virtual reality driving simulator.
 Courtesy of International
 Conference on Virtual Reality.



Head-mounted
 display.
 Courtesy of Journal of
 Virtual Reality.

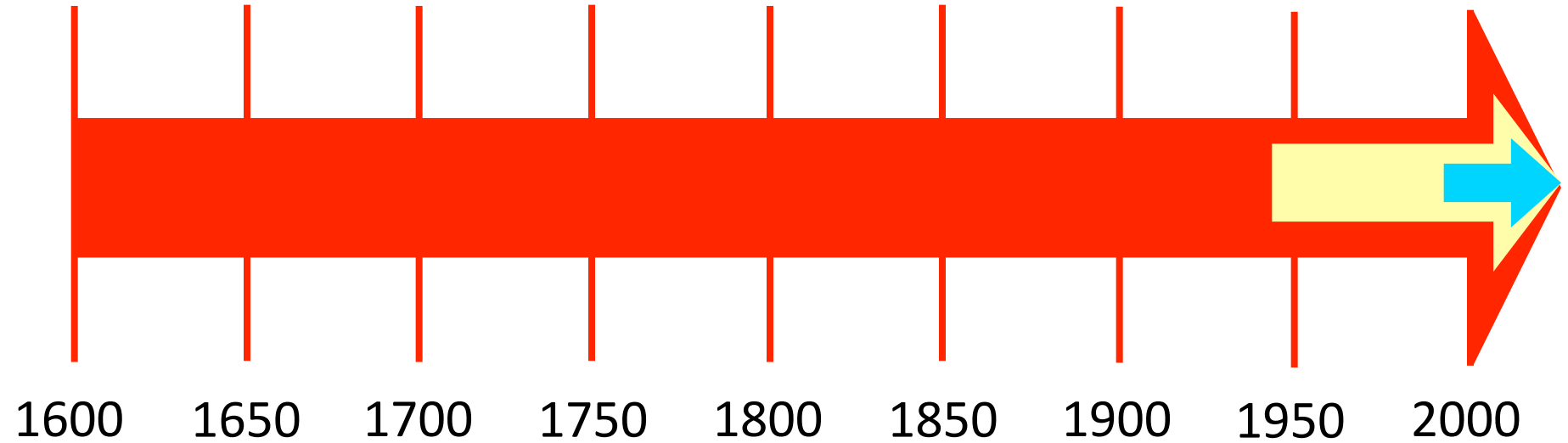


Head-mounted eye tracker.
 Courtesy of EL-MAR Inc.

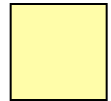


<http://www.youtube.com/watch?v=ABeAAHF6k1k>

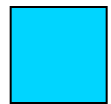
Historical Perspective



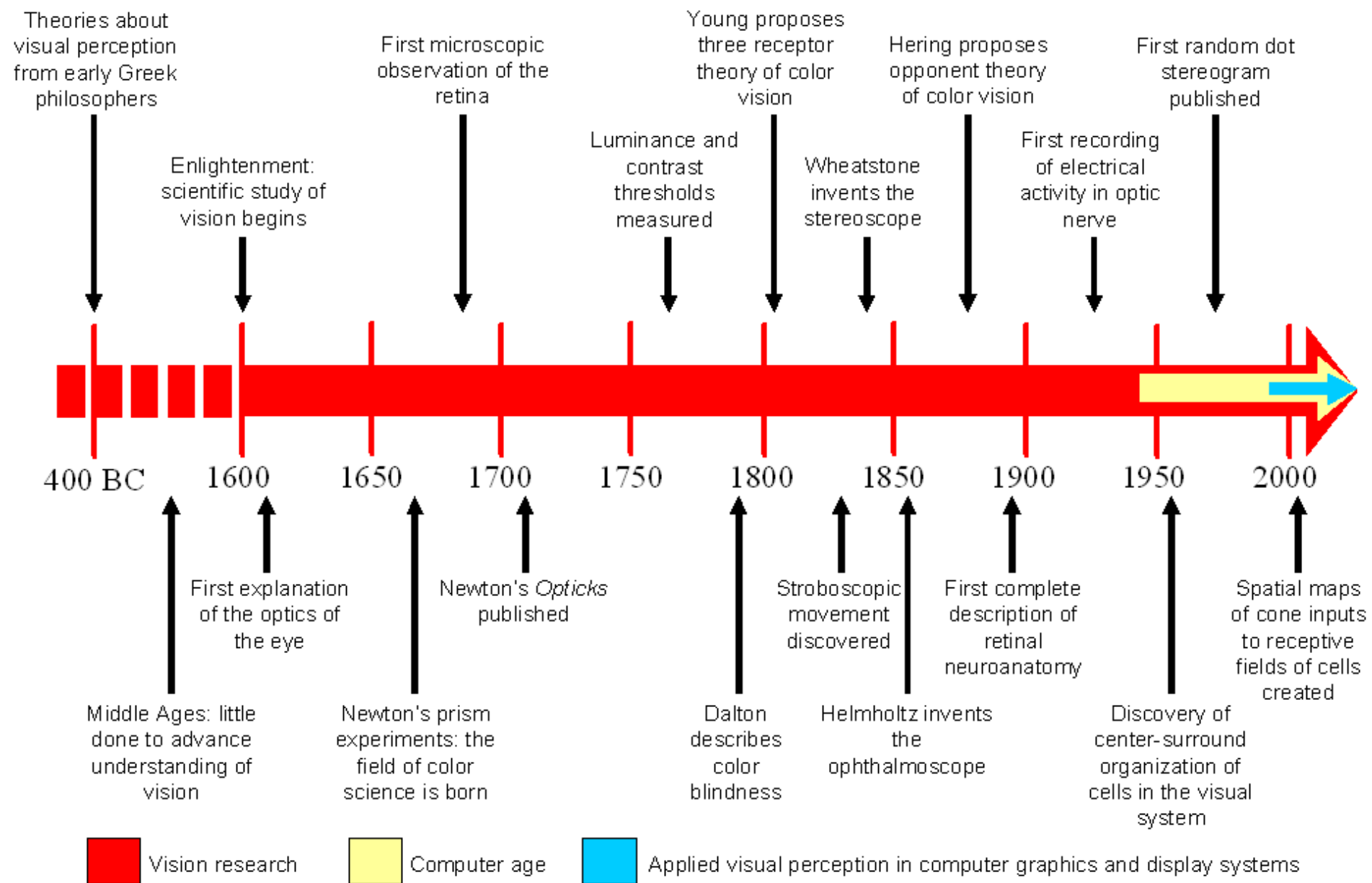
Vision research



Computer age



Applied visual perception in computer graphics



Further Reading

- Computer graphics historical timeline:
 - <http://design.osu.edu/carlson/history/timeline.html>
- Important dates in vision science 1600-1960:
 - http://www.arts.rpi.edu/~ruiz/stereo_history/text/visionsc.html
- Visual science before 1600:
 - http://www.cogsci.uci.edu/vision/yellott_1600.html
- The history of color vision science:
 - <http://www.psych.ucalgary.ca/PACE/VA-Lab/Brian/history.htm>

Research Papers - Class Presentations

- Where do I find papers?
 - <http://www.apgv.org/> (look at the archives)
 - <http://tap.acm.org/>
 - Perception related papers from SIGGRAPH
 - Perception related papers from SIGCHI
 - Eye Tracking Research and Applications Conferences
 - <http://visionscience.com/>
 - Look for links to journals and conferences
 - Augmented reality and virtual reality conferences
 - Do a web search for topics that interest you

Research Papers - Class Presentations

- 20113 Presentation topics
 - Feel free to revisit some of these if you are interested
- 20123 Presentation topics

Possible venues

- 20123 Possible paper venues

Ideas to motivate you as you think about projects

- <http://www.disneyresearch.com/project/teslatouch/>
- <http://www.disneyresearch.com/project/revel-programming-the-sense-of-touch/>
- <http://artimes.rouli.net/2010/07/nothing-like-smell-of-freshly-augmented.html>
- <http://vimeo.com/51920182>
- <http://www.youtube.com/watch?v=8QocWsWd7fc>
- <http://www.youtube.com/watch?v=Jd3-eiid-Uw>
- <http://www.youtube.com/watch?v=bBQQEcfkHoE>